

Geometry Processing with Neural Fields

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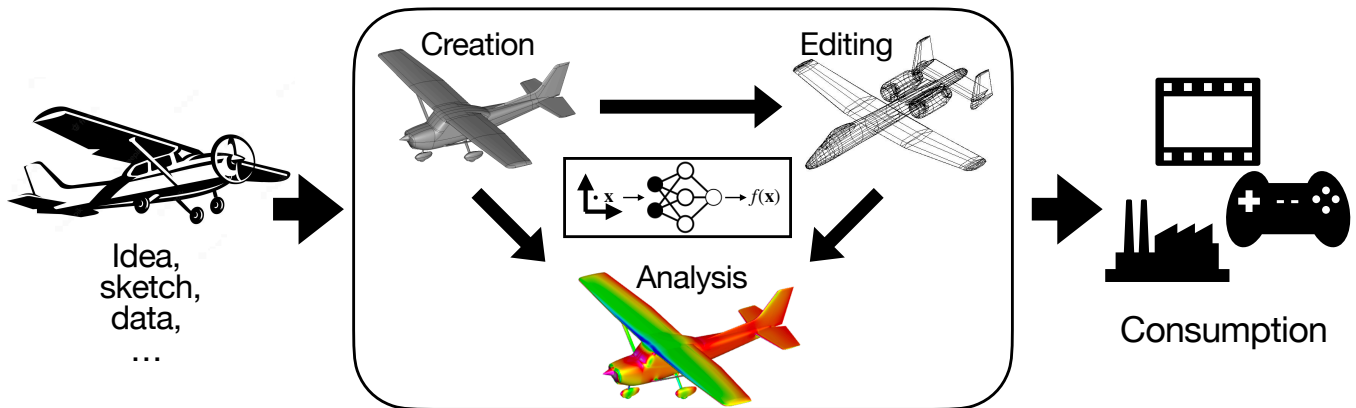


Figure 1: Illustration of a geometry processing pipeline using neural fields as representation.

KEYWORDS

Geometry Processing, Neural Fields, Machine Learning, Deep Learning, Generative Model

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1 INTRODUCTION

Geometry processing is a study of automatic algorithms to handle shapes, including shape creation, manipulation, and analysis. These algorithms are cornerstones for the creation of almost every digital shape. As digital shapes are consumed in our everyday lives, these algorithms are indispensable parts of our every lives.

Traditionally, geometry processing algorithms are usually done with shapes represented explicitly in meshes, voxels, and point clouds. These shape representations are easy to interpret by humans, but they are difficult to be handled by automatic algorithms. For example, voxels can take up lots of memory and meshes might need careful handling of the discretization quality during the processing. These challenges make the development of geometry processing algorithms difficult and inefficient.

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In my thesis, we propose to use an alternative shape representation for the geometry processing pipeline. Specifically, each shape is represented by a neural network that takes a spatial coordinate as input and outputs a scalar or vector value. We coin such representation **neural fields**. Neural field can circumvent the common challenges of meshes and voxels because it doesn't explicitly discretize the space and can be stored compactly. Moreover, it has the unique advantage of being easy to optimize in a deep learning framework, which makes it suitable for data-driven methods. We hypothesize that such a geometry processing pipeline can potentially democratize and revolutionize the digital shape creation paradigm.

Building geometry processing algorithms using neural fields as representation, however, is non-trivial. First, many geometry processing tasks are under-constrained problems. For instance, when creating a shape from scanned data, the point clouds obtained from the scan are often sparse, lacking sufficient information to fully reconstruct the shape. Hence, incorporating *prior knowledge* becomes essential to filter out undesirable solutions. Given that our representation now involves a neural network, encoding such prior knowledge requires a new paradigm. Secondly, neural-field-based algorithms usually rely on stochastic gradient descent optimization, which typically requires numerous iterations to converge. However, this poses a challenge for many geometry processing algorithms that require instantaneous feedback due to user demands. Third, since neural networks can be very complicated non-linear mappings, analyzing them theoretically and deriving provable guarantees is difficult. This limitation poses significant obstacles in application areas where guarantees are needed.

In my thesis, my collaborators and I contribute to tackling these challenges. To tackle the under-constrained problem, we proposed to encode prior knowledge in deep generative models (Section 3)

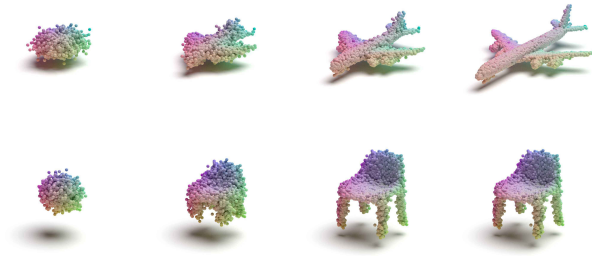


Figure 2: PointFlow’s point cloud transformation.

and in a differentiable loss function (Section 4). These different methods allow us to encode both data-encoded knowledge and physics-inspired knowledge. We found that neural fields can be used as a cache to improve the efficiency of the PDE solvers, which are frequently used to analyze shapes (Section 5). We also propose a class of neural fields called polynomial neural fields, which not only retain the merits of neural fields in general but also can be theoretically analyzed (Section 2). In this paper, I will briefly summarize each of these contributions and end with future work directions.

2 REPRESENTATION - PNF

Neural fields have emerged as a new paradigm for representing signals including image colors [Chen et al. 2021; Sitzmann et al. 2020a], 3D signed distance functions [Atzmon and Lipman 2020; Park et al. 2019a], or radiance fields [Mildenhall et al. 2020]. Recent works have shown that such representations are compact [Dupont et al. 2021; Martel et al. 2021; Saragadam et al. 2022], allow sampling at arbitrary locations [Sitzmann et al. 2020a; Tancik et al. 2020a], and are easy to optimize within the deep learning framework. These advantages enabled their success in many spatial visual computing applications including novel view synthesis [Mildenhall et al. 2020; Niemeyer et al. 2020; Oechsle et al. 2019; Sitzmann et al. 2019; Yariv et al. 2020] and 3D reconstruction [Eslami et al. 2018; Jiang et al. 2020; Peng et al. 2020; Sitzmann et al. 2020a, 2019]. Most recent neural field based methods, however, treat the network as a black box. As such, one can only obtain information by querying it with spatial coordinates. This precludes the applications where we want to change the signal represented in a neural field. For example, it is difficult to remove high frequency noise or change the stationary texture of an image.

One way to enable signal manipulation is to decompose a signal into a number of interpretable components and use these components for manipulation. A general approach that works across a wide range of signals is to decompose them into frequency subbands [Adelson et al. 1984; Burt and Adelson 1987; Simoncelli and Freeman 1995]. Such decompositions are studied in the traditional signal processing literature, e.g., the use of Fourier or Wavelet transforms of the spatial signal. These transformations, however, usually assume the signal is densely sampled in a regular grid. As a result, it is non-trivial to generalize these approaches to irregular data (e.g., point clouds). Another shortcoming of these transforms is that they require a lot of terms to represent a signal faithfully, which makes it difficult to scale to signals of more than two dimensions,

such as in the case of light fields. Interestingly, these are the very problems that can be solved by neural fields, which can represent irregular signals compactly and scale easily to higher dimensions. This leads to the central question of this paper: can we incorporate the interpretability and controllability of classical signal processing pipelines to neural fields?

Our goal is to design a new class of neural fields that allow for precise subband decomposition and manipulation as required by the downstream tasks. To achieve that, our network needs to have the ability to control different parts of the network to output signal that is limited by desirable subbands. At the same time, we want the network to inherit the usual advantages of neural fields, namely, being compact, expressive, and easy to optimize. The most relevant prior works with related aims are Multiplicative Filter Network (MFN) [Fathony et al. 2021] and BACON [Lindell et al. 2022]. While MFNs enjoy the advantages of neural fields and are easy to analyze, they do not use this property to control the network’s output for subband decomposition. BACON [Lindell et al. 2022] extends the MFN architecture to enforce that its outputs are upper-band limited, but it lacks the ability to provide subband control beyond upper band limits. This hinders BACON’s applicability to tasks that requires more precise control of subbands, such as manipulating stationary textures.

To address these issues, we propose a novel class of neural fields called **polynomial neural fields (PNFs)**. PNF is a polynomial neural network [Chrysos et al. 2019] evaluated using a set of basis functions. PNFs are compact, easy to optimize, and can be sampled at arbitrary locations as with general neural fields. Moreover, PNFs enjoy interpretability and controllability of signal processing methods. We provide a theoretical framework to analyze the output of PNFs. We use this framework to design the Fourier PNF, whose outputs can be localized in the frequency domain with both upper and lower band limits, along with orientation specificity. To the best of our knowledge, this is the first neural field architecture that can achieve such a fine-grained decomposition of a signal. Empirically, we demonstrate that Fourier PNFs can achieve subband decomposition while reaching state-of-the-art performance for signal representation tasks. Finally, we demonstrate the utility of Fourier PNFs in signal manipulation tasks such as texture transfer and scale-space interpolation.

3 GENERATION - POINTFLOW

Point clouds are becoming popular as a 3D representation because they can capture a much higher resolution than voxel grids and are a stepping stone to more sophisticated representations such as meshes. Learning a generative model of point clouds could benefit a wide range of point cloud synthesis tasks such as reconstruction and super-resolution, by providing a better *prior* of point clouds. However, a major roadblock in generating point clouds is the complexity of the space of point clouds. A cloud of points corresponding to a chair is best thought of as samples from a distribution that corresponds to the surface of the chair, and the chair itself is best thought of as a sample from a distribution of chair shapes. As a result, in order to generate a chair according to this formulation, we need to characterize a *distribution of distributions*, which is under-explored by existing generative models.

In this paper, we propose PointFlow, a principled generative model for 3D point clouds that learns a distribution of distributions: the former being the distribution of shapes and the latter being the distribution of points given a shape. Our key insight is that instead of directly parametrizing the distribution of points in a shape, we model this distribution as an *invertible parameterized transformation* of 3D points from a prior distribution (e.g., a 3D Gaussian). Intuitively, under this model, generating points for a given shape involves sampling points from a generic Gaussian prior, and then *moving* them according to this parameterized transformation to their new location in the target shape, as illustrated in Figure 2. In this formulation, a given shape is then simply the variable that parametrizes such transformation, and a category is simply a distribution of this variable. Interestingly, we find that representing this distribution too as a transformation of a prior distribution leads to a more expressive model of shapes. In particular, we use the recently proposed continuous normalizing flow framework to model both kinds of transformations [Chen et al. 2018; Grathwohl et al. 2019; Rezende and Mohamed 2015].

This parameterization confers several advantages. The invertibility of these transformations allows us to not just sample but also estimate probability densities. The ability to estimate probability densities in turn allows us to train these models in a principled manner using the variational inference framework [Kingma and Welling 2014], where we maximize a variational lower bound on the log-likelihood of a training set of point clouds. This probabilistic framework for training further lets us avoid the complexities of training GANs or hand-crafting good distance metrics for measuring the difference between two sets of points. Experiments show that PointFlow outperforms previous state-of-the-art generative models of point clouds, and achieves compelling results in point cloud reconstruction and unsupervised feature learning.

4 MANIPULATION - NFGP

In many graphics applications, users may want to edit digital shapes using just a few clicks, such as making a character bow by dragging the head downwards. Such manipulation from sparse input requires geometry processing algorithms. Most of these algorithms use polygonal meshes to represent shapes [Botsch et al. 2010]. Polygonal meshes were created initially for researchers as a representation of real-world shapes that they can both interpret and manipulate [Catmull 1972; Newell 1975; Torchinsky 2013]. However, developing algorithms that automatically manipulate meshes is often difficult since they involve discretizing the surface. For example, changing the topology of a mesh, such as turning a sphere into a torus, will break such discretization and require additional repair procedures such as remeshing [Alliez et al. 2003, 2002]. Given that shape editing is increasingly performed by algorithms, it seems worthwhile to search for a shape representation that is more amenable to automatic geometry processing.

An alternative to the polygonal mesh is an implicit representation, in which the surface is represented by a level set of a field: $\{x | f(x) = c\}$ [Curless and Levoy 1996; Jones et al. 2006; Rosenfeld and Pfaltz 1966]. Since it is easy to change shape topology using implicit representations, people have applied them for geometry processing tasks such as shape merging [Bloomberg and Wyvill

1997; Museth et al. 2002; Osher and Fedkiw 2003]. These works store implicit fields using voxels or octrees, which introduce memory-intensive spatial discretization. Recent research addresses this by using continuous neural networks to represent implicit fields [Chen and Zhang 2019; Mescheder et al. 2019; Michalkiewicz et al. 2019; Park et al. 2019b]. These *neural fields* possess several advantages in addition to the merits inherited from implicit representation: they are compact to store [Davies et al. 2020; Mescheder et al. 2019] and can produce high-quality continuous surfaces at arbitrary resolutions [Mescheder et al. 2019; Park et al. 2019b]. The community has achieved compelling results using neural fields in a variety of applications [Mildenhall et al. 2020; Saito et al. 2019; Takikawa et al. 2021; Yariv et al. 2020]. Notwithstanding these results, it is still unclear whether shape editing tasks challenging for implicit fields (e.g., deformation and filtering) can be performed with neural fields. In this paper, we ask: *can geometry processing be done entirely using neural fields?*

Geometry processing tasks are challenging because they are typically under-constrained: user input is very sparse. Therefore, we need to leverage priors that characterize how a natural surface behaves. Such surface priors are usually instantiated by minimizing energy functionals that describe the physical properties of surfaces using operators from differential geometry. For example, one can encourage surfaces to be smooth by minimizing the curvature, which can be measured by the Laplace-Beltrami operator. Such operators, however, require a parameterization of the surface, which is not available for level sets of a neural field. Mesh algorithms usually approximate these geometric operators using the geodesic neighbors of the surface point. This allows the algorithms to focus on intrinsic geometric properties that are independent of shape parameterization. However, it is not easy to obtain geodesic neighbors in a neural field since the surface of interest is encoded implicitly by a set of points that evaluate the field function to the same value.

We posit that computing these geometric operators in neural fields requires a fundamentally different approach. The idea is to approximate the local surface of the level set using the derivatives of the underlying field. We can evaluate intrinsic geometry properties of the level set, such as curvature, using only the field derivatives [Jones et al. 2006; Osher and Fedkiw 2003]. This allows us to develop loss functions that describe surface priors such as elasticity or rigidity. To achieve this, we use the fact that neural fields are designed to be infinitely differentiable [Sitzmann et al. 2020b; Tancik et al. 2020b]. The infinite differentiability of neural fields makes it possible to optimize loss functions that involve higher-order derivatives using gradient descent methods. Thus, unlike mesh-based geometry processing algorithms that approximate these objectives using surface discretizations, we can directly optimize in terms of the derivatives of the field.

Our formulation provides a proof of concept that geometry processing can be done entirely using neural fields. In particular, we focus on two tasks: shape filtering (e.g., smoothing and sharpening) and topology-preserving deformations. These two tasks are not only essential for downstream applications but also bring out the known challenges associated with implicit representations. We first tackle shape sharpening and smoothing and show how these can be done by optimizing a loss function based on geometric properties

computed via the neural field’s derivatives. To achieve shape deformation, we propose to warp the neural field using a deformation vector field modeled by an invertible neural network [Behrmann et al. 2019]. With this invertible deformation field, we derive a training objective that models the implicitly represented surfaces as elastic shells.

We compare our method with well-established mesh-based baselines [Sorkine-Hornung and Alexa 2007; Taubin 1995]. Our method can match the quality of the mesh-based counterparts without discretizing the surface. We hope that our work can inspire future generations of geometry processing algorithms using neural fields.

5 ANALYSIS - MCGPNF

Solving elliptic PDEs is critical for various computer graphics applications, including 3D reconstruction, animation, and physics simulation. Conventional PDE solvers, however, typically involve time-consuming and error-prone discretization of space with finite elements or meshes. Monte Carlo PDE solvers based on the Walk on Spheres (WoS) algorithm [Sawhney and Crane 2020; Sawhney et al. 2023, 2022] offer a way to circumvent these issues by estimating solution values without discretization. These solvers, however, suffer from high variance, making them slow as they require numerous samples to reduce the variance. This prevents their use in many applications with limited computing budgets.

An alternative to both discretized and Monte Carlo solvers is to use neural fields to approximate the solution to a PDE. Neural fields are a class of neural networks that take spatial coordinates as input and output values of a continuous field [Raissi et al. 2019; Xie et al. 2022]. Prior works have developed self-supervised losses that can be used to optimize a neural field so that it satisfies a given PDE and boundary conditions [Raissi et al. 2019; Sitzmann et al. 2020a]. Neural fields are typically compact, fast to evaluate, and expressive. But training neural fields with such self-supervised losses can be unstable, and they tend to produce biased solutions. Since neural field-based solvers and MC-based solvers have complementary properties, it is natural to ask whether a hybrid solver that combines these two methods can be developed to achieve controllable bias, low variance, and fast evaluation.

This paper takes the first step towards building a hybrid solver combining neural fields with WoS approaches for variable-coefficient elliptic PDEs. Inspired by previous work [Müller et al. 2021] that deploys neural radiance fields as a cache to accelerate Monte Carlo rendering, we hypothesize that a neural field cache can also be used to reduce the variance of a Monte Carlo PDE solver. Following this idea we have developed a novel hybrid PDE solver that first trains a neural field, then uses it to decrease the cost and variance of evaluating the solution. In the training phase, we optimize the neural field supervised by unbiased solution estimates from the WoS algorithm. Then to evaluate the solution, we run WoS but terminate its random walks at a prescribed depth by querying the neural field, providing solution estimates that are more accurate than the neural field alone and faster and less noisy than WoS alone.

We provide a theoretical analysis showing that our loss retains comparable convergence guarantees to conventional SGD algorithms. In practical testing, we find that our neural field has a lower average error than the unbiased but noisy one-sample WoS estimator, and as the number of samples increases, our hybrid solver produces a lower error when the depth limit is set appropriately.

6 CONCLUSION

In this thesis, we propose a novel geometry processing pipeline based on the representation of neural fields using neural networks. While the contributions presented in this thesis, along with ongoing efforts from the research community, highlight the promising potential of neural fields as a powerful representation for geometry processing algorithms, the absence of an end-to-end geometry processing pipeline built entirely on neural fields suggests that there are still intriguing research questions in this area. Here, I outline a few of these questions. First, the effective integration of different geometry processing algorithms within a pipeline requires careful consideration of their compatibility. Second, existing neural fields often operate on functions of spatial points, but many physical fields are functions of spatial volumes. Third, a crucial aspect of the advantage offered by neural fields is their ability to access gradients of arbitrary order through automatic differentiation. However, the usefulness of higher-order gradients obtained analytically via automatic differentiation remains uncertain in practical applications.

In conclusion, this thesis demonstrates that enhancing the efficiency, robustness, and power of the geometry processing pipeline through the use of neural network representation is a highly impactful and promising direction. The novel neural network-based solution has shown great promise in both research and applications, and there is ample evidence to suggest that further success can be achieved through continued efforts within the community.

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